DATA ANALYSIS WITH PANDAS REPORT

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Domain :

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1. Objective

To demonstrate data analysis skills using Pandas by loading a CSV dataset, cleaning data, applying filtering, grouping, and aggregation techniques, and generating meaningful insights.

2. Requirements Covered

- Load and inspect a CSV dataset
- Clean missing or incorrect data
- Apply filtering techniques
- Perform grouping and aggregation
- Generate insights from the dataset
- Use Pandas for data analysis

3. Development Process (Step by Step)

Step 1 - Import Pandas Library

import pandas as pd

Step 2 – Load CSV Dataset

df = pd.read_csv("sales_data.csv")

response = requests.get(url)

Step 3 – Inspect Dataset

print(df.head())

print(df.info())

Step 4 – Clean Missing Values

df.dropna(inplace=True)

Step 5 – Filter Data

electronics = df[df["Category"] == "Electronics"]

Step 6 – Group and Aggregate

```
sales_summary = df.groupby("Category")["Sales"].sum()
```

Step 7 – Display insights

```
print(sales_summary)
```

4. Final Python Code

```
import pandas as pd

# Load CSV file

df = pd.read_csv("sales_data.csv")

print("=== Original Dataset ===")

print(df)

# Inspect dataset

print("\n=== Dataset Information ===")

print(df.info())

# Check missing values

print("\n=== Missing Values ===")

print(df.isnull().sum())

# Filter Electronics products

electronics = df[df["Category"] == "Electronics"]

print("\n=== Electronics Products ===")

print(electronics)

# Group by Category and calculate total sales

category_sales = df.groupby("Category")["Sales"].sum()
```

```
print("\n=== Total Sales by Category ===")

print(category_sales)

# Highest selling product

top_product = df.loc[df["Sales"].idxmax()]

print("\n=== Highest Selling Product ===")

print(top_product)

# Average sales

avg_sales = df["Sales"].mean()

print("\nAverage Sales:", avg_sales)import requests
```

5. Screenshots with Captions

Figure 1 : Figure 1: Pandas Library Installation

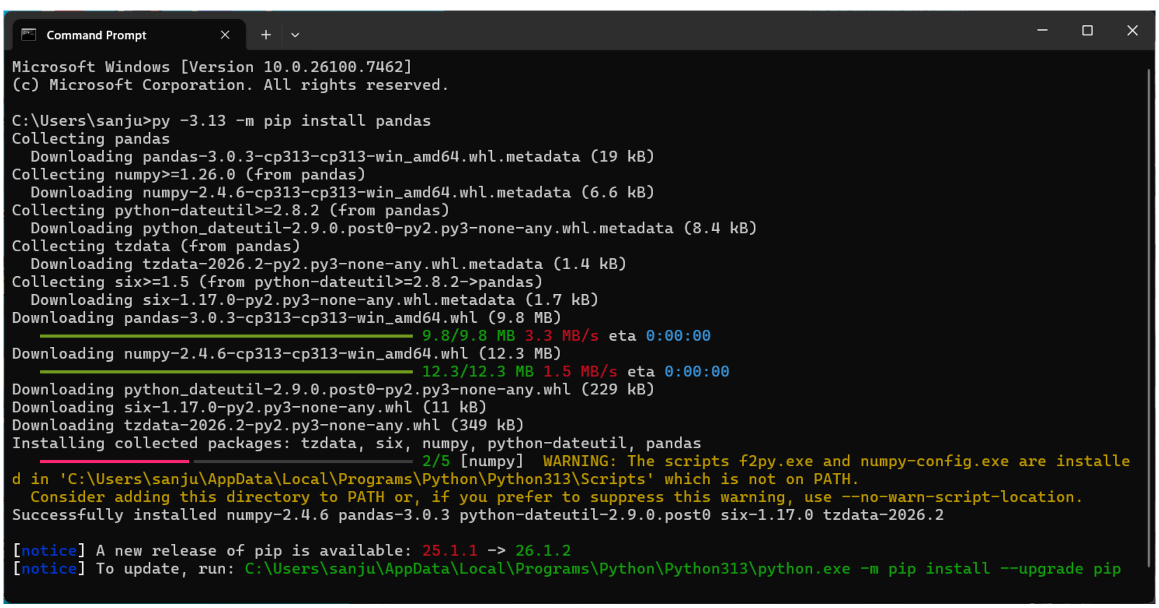
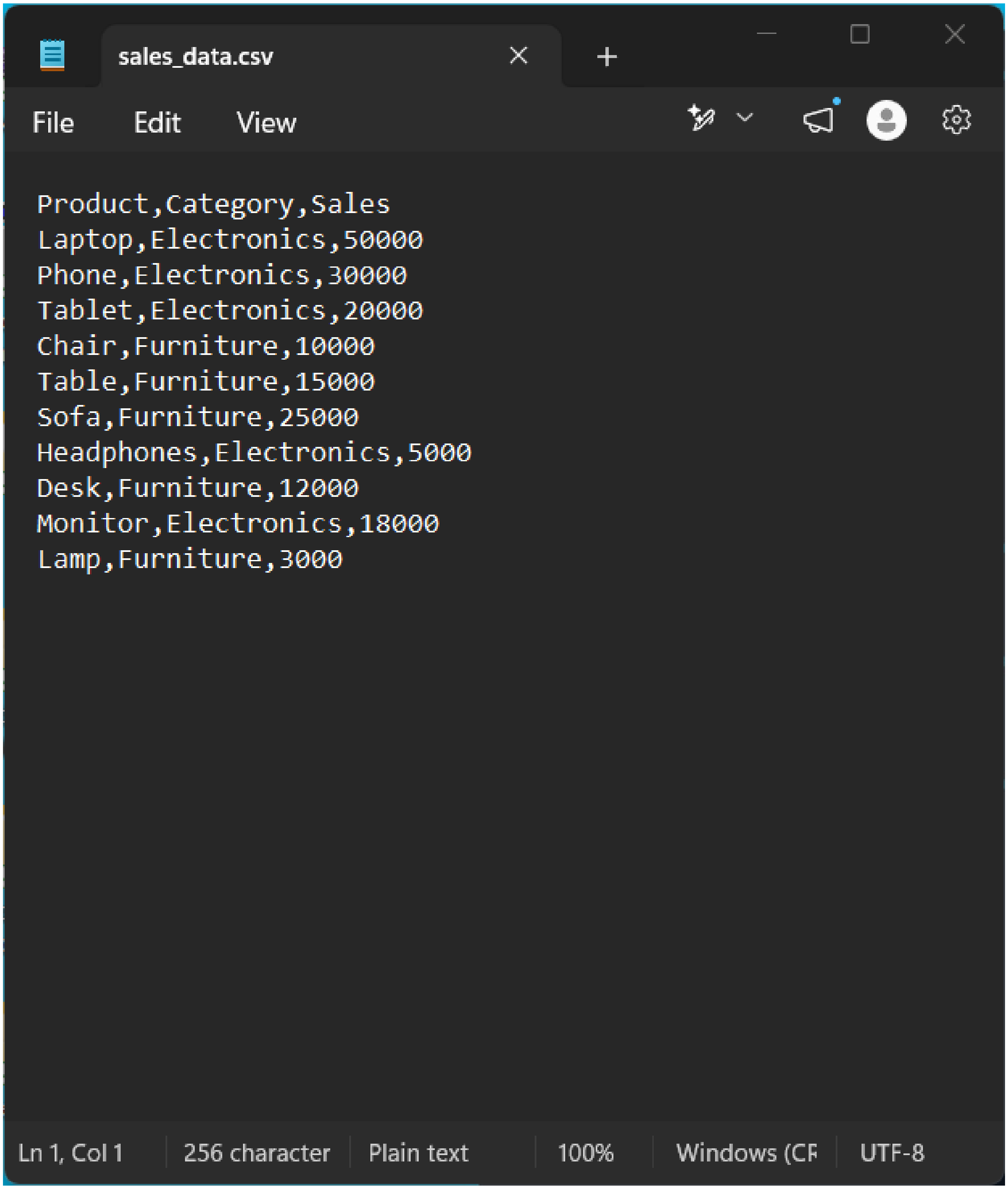


Figure 2 : Figure 2: Sales Dataset Creation (sales_data.csv




```
Product,Category,Sales
Laptop,Electronics,50000
Phone,Electronics,30000
Tablet,Electronics,20000
Chair,Furniture,10000
Table,Furniture,15000
Sofa,Furniture,25000
Headphones,Electronics,5000
Desk,Furniture,12000
Monitor,Electronics,18000
Lamp,Furniture,3000
```

Figure 3 : Figure 3: Pandas Data Analysis Program

task3_pandas.py - C:/Users/sanju/OneDrive/Desktop/task3_pandas.py (3.13.5)

File Edit Format Run Options Window Help

```python
import pandas as pd

# Load CSV file
df = pd.read_csv("sales_data.csv")

print("=== Original Dataset ===")
print(df)

# Inspect dataset
print("\n=== Dataset Information ===")
print(df.info())

# Check missing values
print("\n=== Missing Values ===")
print(df.isnull().sum())

# Filter Electronics products
electronics = df[df["Category"] == "Electronics"]

print("\n=== Electronics Products ===")
print(electronics)

# Group by Category and calculate total sales
category_sales = df.groupby("Category")["Sales"].sum()

print("\n=== Total Sales by Category ===")
print(category_sales)

# Highest selling product
top_product = df.loc[df["Sales"].idxmax()]

print("\n=== Highest Selling Product ===")
print(top_product)

# Average sales
avg_sales = df["Sales"].mean()

print("\nAverage Sales:", avg_sales)
```

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Figure 4 : Dataset Inspection Using Pandas

```
IDLE Shell 3.13.5
File  Edit  Shell  Debug  Options  Window  Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=========== RESTART: C:\Users\sanju\OneDrive\Desktop\task3_pandas.py ===========
=== Original Dataset ===
      Product     Category  Sales
0      Laptop  Electronics  50000
1       Phone  Electronics  30000
2      Tablet  Electronics  20000
3       Chair    Furniture  10000
4       Table    Furniture  15000
5        Sofa    Furniture  25000
6  Headphones  Electronics   5000
7        Desk    Furniture  12000
8     Monitor  Electronics  18000
9        Lamp    Furniture   3000

=== Dataset Information ===
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 3 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Product   10 non-null     str
 1   Category  10 non-null     str
 2   Sales     10 non-null     int64
dtypes: int64(1), str(2)
memory usage: 372.0 bytes
None
```

Figure 5 : Final Data Analysis Output

```
=== Missing Values ===
Product     0
Category    0
Sales       0
dtype: int64

=== Electronics Products ===
      Product     Category  Sales
0      Laptop  Electronics  50000
1       Phone  Electronics  30000
2      Tablet  Electronics  20000
6  Headphones  Electronics   5000
8     Monitor  Electronics  18000

=== Total Sales by Category ===
Category
Electronics    123000
Furniture       65000
Name: Sales, dtype: int64

=== Highest Selling Product ===
Product          Laptop
Category    Electronics
Sales             50000
Name: 0, dtype: object

Average Sales: 18800.0
>>>
```

6. Insight Summary

The sales dataset was successfully analyzed using the Pandas library. The dataset contained 10 records with no missing values, indicating good data quality. After filtering the Electronics category, five electronic products were identified. Category-wise aggregation revealed that Electronics generated the highest total sales of **123,000**, while Furniture recorded total sales of **65,000**. The **Laptop** was identified as the highest-selling product with sales of **50,000**, and the average sales value across all products was **18,800**. These insights demonstrate how Pandas can be used to extract meaningful information and support data-driven decision-making.

7. Conclusion

This project successfully demonstrated data analysis using the Pandas library. The CSV dataset was loaded, inspected, and analyzed through filtering, grouping, and aggregation techniques. The analysis helped identify category-wise sales performance, the highest-selling product, and average sales values. The project fulfilled all the requirements and provided practical experience in data analysis and insight generation using Python.